APM348 Mathematical Modelling – Midterm 1

University of Toronto
Faculty of Arts & Science

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65 Minutes

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Solution

Exam Reminders:

- **Keep this booklet closed**. Do not begin writing the exam until instructed to do so. You may fill out your information above before the test begins.
- Please place your **student ID card** on your desk.
- As a student, you help create a fair and inclusive writing environment; unauthorized aids are prohibited and using one may result in you being charged with an academic offense.
- **No aids are allowed**: no calculators, no textbook, no notes, and no assistance.
- **Turn off** and place all cell phones, smart watches, electronic devices, and unauthorized study materials **in your bag under your desk**. These devices may not be left in your pockets.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an exam facilitator.
- In the event of a fire alarm, do not check cell phones or other electronic devices unless authorized to do so.

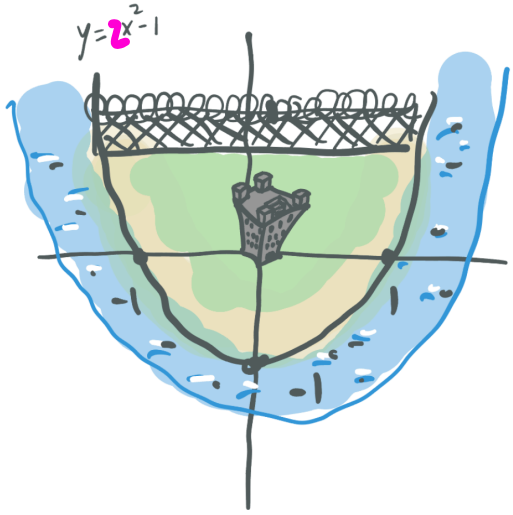
Exam Writing:

- Please write your answers **in the boxes**. There is ample space within each one. If you must use additional space, please use the blank pages at the end of this booklet and clearly indicate in the given box that your answer is **continued on page** [number]. You can also use the blank pages as scrap paper.
- This booklet contains 8 pages. If your booklet is missing a page, please raise your hand to notify an exam facilitator.
- **Do not remove or unstaple any page from this booklet.**
- Remember to show all your work and explain your answers using full English sentences.

Good luck!

Question:	**1**	**2**	**Total**
Points:	26	24	**50**

1 A prisoner is trying to escape and wants to find the furthest spot from the main building on the prison compound to plot his escape!



The main building is located at the origin.

The bank of the river on the prison compound is given by the function $y = 2x^2 - 1$ and the North fence is located at $y = \frac{1}{2}$.

(a) (6 points) Set up the prisoner's goal as an optimization problem.

The distance between the prisoner's position (x,y) and the origin is $\sqrt{x^2+y^2}$, so we want to maximize this quantity.

The prisoner is confined to the prison compound, so $2x^2-1 \le y \le 1/2$.

Optimization problem:

$$\max \sqrt{x^2+y^2}$$
$$\text{subject to } 2x^2-1 \le y \le 1/2$$

It is equivalent to consider the square of the distance, since its maximum is attained for the same values of (x,y).

So another possible solution is:

$$\max (x^2+y^2)$$
$$\text{subject to } 2x^2-1 \le y \le 1/2$$

(b) (10 points) Solve the optimization problem. Justify your answer.

Let $f(x,y) = x^2 + y^2$, the optimization function.
The maximum can be either in the interior or on the boundary.

Interior If this is the case, then $\nabla f(x,y) = (0,0)$.
Thus $(2x, 2y) = (0,0) \iff x = y = 0$ which gives $f(0,0) = 0$ ✗ this is a min

Boundary: North Fence In this case $y = 1/2$, so we are optimizing:

$$\max \underbrace{\left(x^2 + 1/4\right)}_{g(x)} \quad \text{for } \frac{-\sqrt{3}}{2} < x < \frac{\sqrt{3}}{2}$$

we find the endpoints where the fence meets the river:
$1/2 = 2x^2 - 1$.

Then $g'(x) = 0 \iff 2x = 0 \iff x = 0$.
But $g(0) = 1/4$ ✗ is the min on this boundary.

Boundary: River Here $y = 2x^2 - 1 \iff x^2 = \frac{y+1}{2}$, so we are solving

$$\max \underbrace{\left(\frac{y+1}{2} + y^2\right)}_{h(y)}$$
$$-1 \le y \le 1/2$$

we check $-1 < y < 1/2$: $h'(y) = 0 \iff \frac{1}{2} + 2y = 0 \iff y = -1$ and we get $\begin{cases} x = 0 \\ h(-1) = 1 \end{cases}$

we also need to check the endpoints:
$y = 1/2$: we get $x = \pm\frac{\sqrt{3}}{2}$ and $h(1/2) = 1$

Coordinates of solution: $(0,-1)$ or $\left(-\frac{\sqrt{3}}{2}, 1/2\right)$ or $\left(\frac{\sqrt{3}}{2}, 1/2\right)$

Distance to the main building: $\sqrt{1} = 1$

(c) (10 points) The rain over the Summer was different from the usual and the river bank changed from $y = 2x^2 - 1$ to $y = 2x^2 - 0.9$. Does the prisoner's optimal distance increase, decrease, or doesn't change? Estimate how much the prisoner's optimal distance changes. Justify your answer.

For this question we need to change the constraint $2x^2 - y = 1$ to $2x^2 - y = 0.9$.

We can go back and redo the problem with the river constraint $2x^2 - y = R$ and see how the maximum changes at the points we found.

River South Pt $x = 0$ and $y = -R$, distance is R

So the distance will change by $\approx \frac{\partial \text{dist}}{\partial R}(0.1)\Big|_{R=1}(0.1) = 0.1$

NE and NW corners $y = 1/2$ and $x = \pm\sqrt{\frac{R+1/2}{2}}$, distance is

$$\sqrt{\frac{R+1/2}{2} + \frac{1}{4}} = \sqrt{\frac{R}{2} + \frac{1}{2}}$$

and the distance will change by approximately

$$\frac{\partial \text{dist}}{\partial R}\Big|_{R=1}(0.1) \approx \frac{1/2}{2\sqrt{R/2+1/2}}\Big|_{R=1}(0.1) = \frac{0.1}{4} = 0.025$$

Also the river bank moved "up" which means that it is "flooding" so the max distance decreased by the smallest change of 0.025.

Optimal distance: increases / **decrease** / doesn't change

Estimation of the optimal distance change: 0.025

Note that if the river changed to $2x^2 - y = 1.1$, then the change would have been 0.1 because we would choose the point where the max increased the most. This makes it tricky to solve using Lagrange multipliers.

2 Suppose a handblown wine glass factory receives an order for G wine glasses to be delivered at time T.

The glasses are fragile, so storing them is costly, so they want to don't want to produce them early. On the other hand, producing the glasses late incurs a larger cost because producing glasses faster is more wasteful.

Assume that initially we don't have any glasses stored.

Wwe define

- $t =$ the amount of time (in days) since the order was placed;
- $x(t) =$ the (accumulated) number of wine glasses produced by time t for this order.

Putting all this together, we obtain the following optimization problem:

$$\min_{x(t)} \int_0^T \underbrace{e^{-t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2}_{F\left(t, x, \frac{dx}{dt}\right)} \mathrm{d}t,$$

where $x(t)$ satisfies certain constraints.

(a) (8 points) What are the constraints satisfied by $x(t)$? Justify your answer.

At the beginning we have nothing produced : $x(0)=0$.
We want to have G glasses at time T : $x(T)=0$.
($x(t)$ also needs to have a differentiable derivative.)

(b) (12 points) Find the optimal production rate $x(t)$ that minimizes the total cost. Justify your answer.

We use the Euler-Lagrange equation : $\frac{d}{dt}\left(\frac{\partial F}{\partial x'}\right) = \frac{\partial F}{\partial x}$.
This yields $\frac{d}{dt}\left(2e^{-t}x'(t)\right) = 0 \iff -2e^{-t}x'(t) + 2e^{-t}x''(t) = 0$
or $x''(t) = x'(t) \Rightarrow x'(t) = x(t) + A \Rightarrow e^{-t}x'(t) - e^{-t}x(t) = Ae^{-t}$
$\Rightarrow \left(e^{-t}x\right)' = Ae^{-t}$. Thus $e^{-t}x(t) = -Ae^{-t} + B \iff x(t) = -A + Be^{t}$
Using our constraints : $x(0)=0$ and $x(T)=G$, we get

$x(t) =$ $G \cdot \frac{e^t - 1}{e^T - 1}$

(c) (4 points) For what values of T and G does the optimal production require the factory to produce negative amounts of glasses? If this occurs, how could the plan be adjusted to ensure the production rate remains nonnegative?Justify your answer.

we need to check when $x'(t) < 0$:

$$x'(t) = \frac{G}{e^T - 1} e^t.$$

Since $T, G > 0$, this is always positive.

This never occurs for the optimal production.